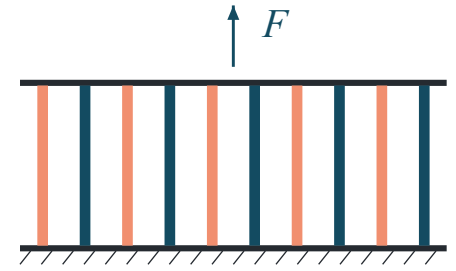
DANIELS SYSTEM



ADVERSARIAL KNOWLEDGE



Knowledge state κ_h

h = 3 known identities

? Unknown type

VALUE OF INFORMATION OF ROBUSTNESS

Most damaging action $a^*(d, \kappa)$

Fixed attack size r

Survival / extinction probabilities

$$I_{\text{rob}} = \exp(-D_{\text{KL}})$$

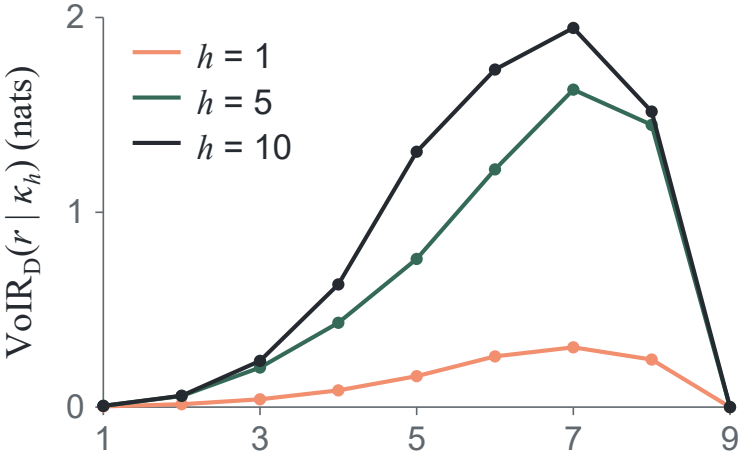
Intact-system reference

$$\text{VoIR}_{\text{D}}(d \mid \kappa \neq \kappa_0) = D^*_{\text{KL}}(d \mid \kappa) - D^*_{\text{KL}}(d \mid \kappa_0)$$

Compared with uninformed attacks

DANIELS EXAMPLE

Selected knowledge states



More knowledge can reduce system robustness. The value of knowledge rises and then falls as attack size increases in this example.